

QLoRA Fine-Tuning of PaliGemma for Remote Sensing Image Captioning

İlayda Ergür

Middle East Technical University, Ankara, Turkey

Abstract—Large Vision-Language Models (VLMs) show strong general-purpose performance but require domain-specific fine-tuning to perform well on specialised tasks such as remote sensing image captioning. In this work, we apply QLoRA [2] — 4-bit quantisation combined with low-rank adapters — to fine-tune PaliGemma-3B [1] on the RISC dataset, training only 0.39% of the model’s parameters. Our approach achieves substantial gains over the pre-trained baseline: BLEU-4 improves from 0.0062 to 0.0553 ($\times 9$), METEOR from 0.0897 to 0.2285, and ROUGE-L from 0.1476 to 0.2675.

I. INTRODUCTION

Remote sensing image captioning is a challenging vision-language task requiring models to describe satellite imagery using precise, domain-specific vocabulary. PaliGemma [1] is a 3B-parameter VLM combining the SigLIP vision encoder with the Gemma language decoder, pre-trained on diverse image-text pairs. Although capable across many benchmarks, it lacks familiarity with satellite imagery terminology and produces generic outputs without adaptation.

QLoRA [2] quantizes the frozen base model weights to 4-bit precision while training a small set of low-rank adapter (LoRA) matrices. This reduces trainable parameters to less than 1% of the total, enabling high-quality fine-tuning on a single GPU.

II. METHOD

A. Dataset

We use the RISC dataset [3] (caglar Mert/full_riscm), comprising 44,521 satellite images at 224×224 px resolution, each annotated with five human-written captions covering diverse remote sensing scenes. The dataset is split 95/5 into 42,294 training image-caption pairs and 2,226 test images. Evaluation is performed on a 200-sample subset using one ground-truth caption per image, as required by the assignment specification.

B. Model and QLoRA Setup

PaliGemma-3B (google/paligemma-3b-pt-224) is loaded in 4-bit NF4 quantisation using BitsAndBytes with bfloat16 compute dtype. LoRA adapters with rank $r=8$ are attached to all seven linear projection layers of the Gemma language decoder: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. The SigLIP vision encoder and the multi-modal projector are frozen throughout, focusing adaptation capacity on language generation. This yields **11.3M trainable parameters out of 2.93B total (0.39%)**, as reported by `model.print_trainable_parameters()`.

Training uses the AdamW-8bit optimiser ($\beta_2 = 0.999$), learning rate 2×10^{-5} , effective batch size 16 (4 per device $\times$ 4 gradient accumulation steps), 100 warm-up steps, and bfloat16 mixed precision for 1 full epoch. Each input is formatted as the text prompt “*describe this image.*” paired with the satellite image; the corresponding caption serves as the generation target. The processor produces input tokens with image embeddings prepended, and cross-entropy loss is computed only on caption tokens.

III. RESULTS AND DISCUSSION

A. Quantitative Evaluation

Table I and Fig. 2 report BLEU-4, METEOR, and ROUGE-L on 200 held-out test images. All three metrics improve consistently after QLoRA fine-tuning. BLEU-4 increases nearly 9 $\times$, from 0.0062 to 0.0553. METEOR and ROUGE-L, which better account for synonym overlap and recall, both roughly double (0.0897→0.2285 and 0.1476→0.2675, respectively). Low absolute BLEU-4 values are typical for single-reference captioning evaluation and are consistent with reported scores on standard benchmarks under the same protocol.

TABLE I: Captioning metrics on 200 test images (single GT reference)

Model	BLEU-4	METEOR	ROUGE-L
Baseline PaliGemma	0.0062	0.0897	0.1476
QLoRA Fine-tuned	0.0553	0.2285	0.2675

B. Qualitative Evaluation

Fig. 1 shows five held-out test images with their ground-truth caption, baseline output, and QLoRA fine-tuned output. The baseline produces generic one- or two-word responses, while the fine-tuned model generates specific, sentence-level descriptions consistent with the RISC training style.

IV. CONCLUSION

We demonstrated that QLoRA enables effective domain adaptation of PaliGemma-3B with only 0.39% trainable parameters, using a single A100 GPU for one training epoch. Fine-tuning on the full RISC dataset yields large quantitative improvements across BLEU-4, METEOR, and ROUGE-L, and produces qualitatively richer, domain-specific captions for remote sensing imagery. These results confirm that parameter-efficient fine-tuning is a viable and practical strategy for adapting large VLMs to specialised vision-language tasks without incurring the cost of full fine-tuning.

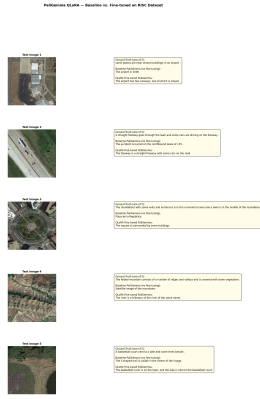


Fig. 1: Five test images with ground truth, baseline PaliGemma, and QLoRA fine-tuned captions.

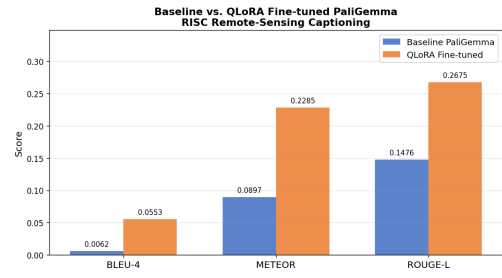


Fig. 2: BLEU-4, METEOR, and ROUGE-L scores for baseline and QLoRA fine-tuned PaliGemma.

REFERENCES

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